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3D-printed concrete with in-process embedded fiber-reinforced polymer grid reinforcement

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The absence of effective reinforcement has hindered the widespread use of 3D-printed concrete. In this study, we propose an innovative in-process embedding technology that enables the simultaneous printing of concrete and flexible fiber-reinforced polymer grids. We investigate the performance of grids as reinforcement in 3D-printed concrete through pull-out and splitting tensile tests to assess the bonding properties between the grids and concrete. Additionally, three-point bending tests were conducted to evaluate the impact of grids on the flexural performance of 3D-printed concrete plates. Test results show that the load-bearing capacity and deflection of 3D-printed concrete plates reinforced with grids increased by approximately 41% and 552%, respectively. However, the voids and reduced contact area caused by the simultaneous printing of the grids weakened the bonding between adjacent concrete layers. Overall, the findings confirm the feasibility of the proposed in-process embedding technology and demonstrate its potential to advance 3D-printed concrete structures through the effective integration of in-process fiber-reinforced polymer reinforcement.

Traditional construction methods face challenges such as environmental pollution, labor shortages, and high energy consumption^{1–4}. To address these, 3D printing technology has been increasingly applied in construction, particularly in 3D-printed concrete (3DPC), which reduces labor costs, shortens construction cycles, and building materials compared to traditional methods⁴. However, as 3DPC applications grow, the challenge of reinforcing 3DPC components has become a major concern^{5–10}. Unlike mould-cast concrete, 3DPC struggles with integrating reinforcement during the printing process due to its layer-by-layer construction. Unreinforced 3DPC components have weak flexural and tensile strength and are prone to brittle failure, limiting the potential of this technology.

To overcome this issue, researchers have proposed various reinforcement methods, including fiber reinforcement, steel reinforcement, and fiber-reinforced polymer (FRP) reinforcement^{11,12}. The fiber reinforcement method, which involves adding a high content of fibers to concrete to print engineering cementitious composites (ECC), noticeably enhances the ductility and strain-hardening capacity of 3DPC components, and enabling fully automated printing^{13–17}. However, to ensure the printability of 3DPC, the fiber content cannot exceed a certain threshold, limiting the application of this method¹⁸. Steel is widely used as reinforcement in mould-cast concrete due to its strong bonding properties with concrete^{19–21}. For this application in 3DPC, it can be categorized into pre-installed reinforcement, in-process continuous

reinforcement, and post-installed reinforcement, depending on the fabrication stage.

Regarding the pre-installed reinforcement, the most common method involves using 3DPC technology to prepare the formwork, placing the steel bars, and then pouring concrete to create an integrated structure^{22–24}. The post-installed reinforcement method involves manually placing steel after the deposition of concrete filaments manually or using an additional machine^{18,25,26}, which is generally complicated. Although these methods successfully reinforce 3DPC structures, the process cannot be fully automated. To address this issue, a series of reinforcement methods implemented simultaneously with concrete printing have been proposed, including simultaneous printing of reinforcement bars or steel cables^{27–29}. The installation of steel reinforcement in 3DPC, however, is not intelligent and introduces additional porosity in concrete^{19,30}. In steel-reinforced concrete structures, excessive porosity can lead to corrosion of the steel by harmful ions, thus compromising the overall service life of structures. Therefore, applying steel to 3DPC may not be an optimal solution¹⁹.

More recently, fiber-reinforced polymer (FRP) reinforcement, which possesses excellent tensile strength and durability^{15,16,19,31–37}, has been adopted as reinforcement for 3DPC structures. The author team proposed enhancing the flexural performance of 3D printed high-performance concrete plates by placing FRP grids and FRP bars between the printed layers. Their results demonstrated that FRP noticeably improves both the flexural

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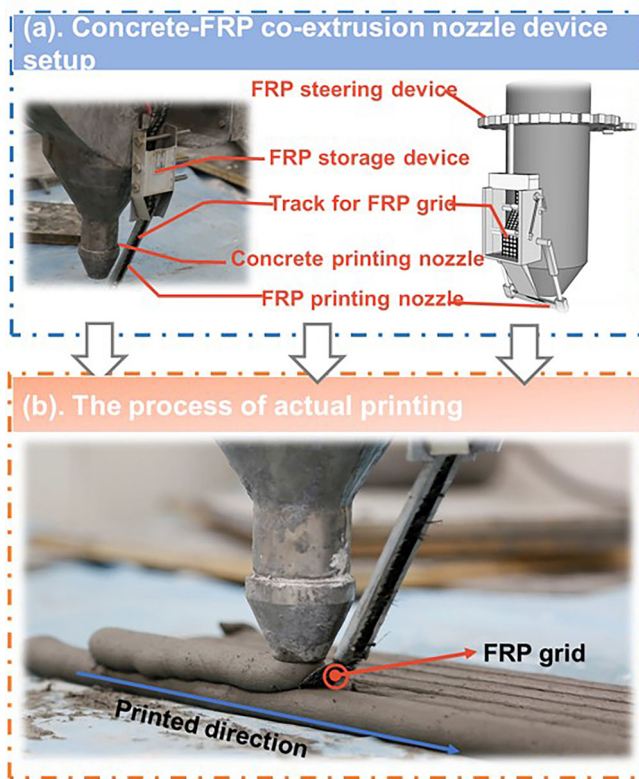


Fig. 1 | Innovative concrete-FRP in-process embedded nozzle. a Concrete-FRP in-process embedded device; **b** Simultaneously printing of FRP reinforcement and concrete.

strength and ductility of 3DPC plates^{16,38}. Similarly, A previous study explored the bond performance between BFRP bars and 3DPC. It was found that the bond strength of the 3D printed specimens was lower than that of the mould-cast specimens, and the damage pattern of the BFRP bar reinforced printed concrete was similar to that of reinforced mould-cast concrete³⁹. Although these studies have highlighted the potential of FRP for reinforcing 3DPC, a key challenge remains: fully automated printing has not yet been achieved, as FRP still needs to be placed manually.

On the other hand, using functional materials for functional graded concrete (FGC) is an innovative approach to optimally use construction materials, which is achieved by introducing different types of materials or additives with a gradient in performance. This design allows FGC to differentiate performance in different parts of a structure to meet specific performance needs⁴⁰. Despite the noticeable advantages of FGC, the existing production process is still immature, which limits its practical application to a certain extent. 3D printing technology can realize the precise preparation of FGC components owing to the layer-by-layer construction technology, which overcomes the limitations of traditional manufacturing methods^{17,40}.

In this study, an innovative device that enables the simultaneous printing of concrete and FRP reinforcement has been developed. This technique adopts a multi-nozzle system (Fig. 1) equipped with a track and storage device for flexible FRP reinforcement. To better understand the performance of FRP-reinforced 3DPC plates and to verify the proposed in-process embedded device, we used the in-process embedded device to fabricate FGC plates composed of flexible FRP grids, geopolymer concrete (GC) and fiber-reinforced concrete (FC), and conducted three-point bending tests. Given the impacts of hydroxypropyl methylcellulose (HPMC) and sodium metasilicate pentahydrate (PSM) on the hydration behavior of FC and GC—which directly govern long-term strength development—we conducted comparative X-ray diffraction (XRD) and thermogravimetric analysis (DTG-TG) to elucidate their effects on hydration product formation in both systems. We explored the bonding performance

between the flexible FRP grid and the concrete, as well as between adjacent printed concrete layers, through pull-out tests and interlayer interface splitting tensile tests. Finally, we conducted mercury intrusion porosimetry (MIP) and micro-computed tomography (μ CT) to the pore structures of the layer-by-layer interface. While this study presents a proof of concept of this multi-nozzle system, the effects of parameters on flexural performance of 3D printed concrete plates with in-process FRP reinforcement have been presented in a separated paper³².

Methods

Innovative concrete-flexible FRP in-process embedded system

The main contribution of our proposed in-process embedded system is that it integrates an additional nozzle for extruding FRP reinforcement alongside the concrete printing nozzle. The concrete nozzle is based on the YX1208-NS 3D concrete printer, manufactured by Hangzhou Guan Li Intelligent Technology Co., Ltd. During the printing process, we extrude concrete through the concrete nozzle, while flexible FRP reinforcement is fed through the FRP nozzle and settles beneath the concrete layer. The gravity of the concrete automatically compacts the FRP grids at the interface of adjacent printed concrete layers (Fig. 1). A key feature of our in-process embedded system is that the FRP nozzle is positioned behind the concrete nozzle during printing.

As shown in Fig. 1, the in-process embedded system consists of an FRP steering device, an FRP storage device, an FRP printing nozzle, and a concrete printing nozzle. This setup enables us to print concrete and flexible FRP reinforcement simultaneously. The development of the in-process embedded nozzle includes the following key elements (Fig. 1): (a) The track for the flexible FRP grids is designed to be easily disassembled, ensuring that FRP grids of different widths can pass smoothly. The FRP printing nozzle is slightly extended below the concrete nozzle, allowing the FRP grids to be embedded at the center of the concrete filaments. (b) The FRP steering and storage devices are connected to the concrete nozzle. The printing procedure was as follows: The FRP was continuously fed from a storage system along a rail to the FRP printing nozzle. Here, it was co-extruded with the concrete filament to concurrently deposit the FRP-reinforced 3DPC structure. A steering mechanism guided the FRP during directional changes of the print path, while the FRP feed rate was governed directly by the concrete printer's software. Noted that the synchronous printing systems proposed by the previous study²⁸ provide a good approach for co-printing of concrete and micro-cables, while our dual-nozzle in-process embedded device incorporates modular, detachable FRP grid tracks and nozzles—enabling adaptation to FRP grids of varying sizes and noticeably enhancing versatility and utilization efficiency. It should be specifically noted that the device developed in this study enables only the in-plane synchronous embedding of FRP grids. It does not permit the placement of reinforcement across interlayer interfaces—that is, oriented orthogonally to the printing direction—which would be essential for enhancing interlayer bonding performance. This limitation points to a critical and yet unresolved challenge in the field of 3DPC. The layer-by-layer nature of the additive manufacturing process results in distinct interlayer weaknesses, noticeably increasing concrete porosity at these interfaces. This heightened porosity facilitates the ingress of deleterious ions, such as chlorides and sulfates, thereby accelerating deterioration and compromising the long-term durability of the structure. Further research to address this issue is therefore imperative and will be pursued in subsequent studies.

Preparation of 3D-printed concrete

We conducted three types of tests: three-point bending tests, splitting tensile tests of the bond interface, and pull-out tests. The specimen design is detailed in the next section. Since the 3DPC uses FGC, we applied two types of concrete: FC and GC. The mix proportions for both concrete types, based on trial tests and confirmed to have good printability, are listed in Supplementary Table 1.

The printer was equipped with a 20 mm diameter nozzle and operated at an extrusion speed of 1.5 cm³/s. The nozzle movement speed was adjusted

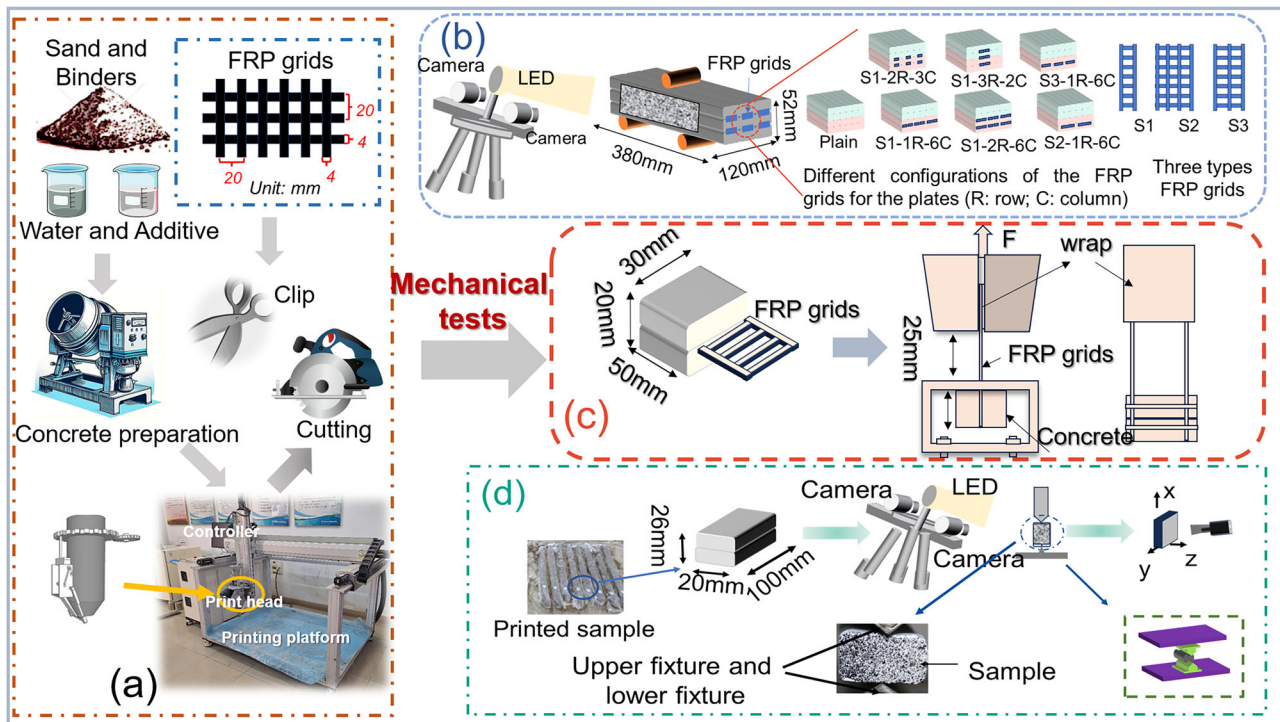


Fig. 2 | Concrete elements with in-process embedded FRP reinforcement and mechanical testing. a FRP grid reinforcement concrete printing; **b** Three-point bending tests; **c** Pull-out tests; **d** Splitting tensile tests.

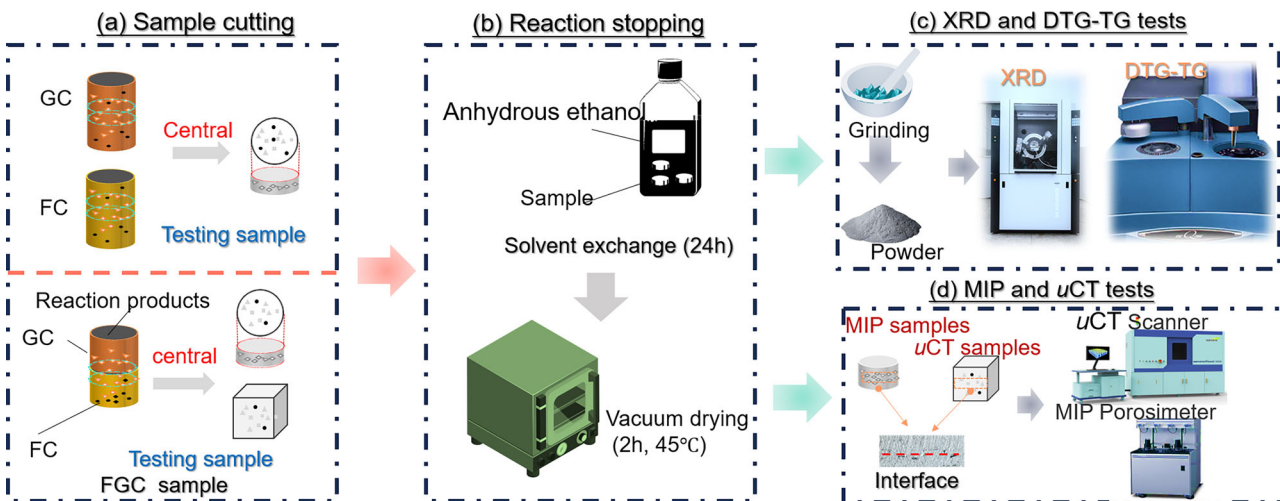


Fig. 3 | Microscopic characterization testing procedure. a Sample cutting. **b** Reaction stopping. **c** XRD and DTG-TG tests. **d** MIP and uCT tests. GC geopolymer concrete, FC fiber-reinforced concrete, FGC functionally graded concrete, MIP mercury intrusion porosimetry, uCT micro-computed tomography, XRD X-ray diffraction, DTG-TG thermogravimetric analysis.

based on FRP grid-concrete compatibility. The interval time between printing FC and GC was set to 25 min, which includes cleaning, refilling, and repositioning the printer. As this paper mainly focuses on the feasibility of our novel dual-nozzle in-process embedded technology for simultaneously printing flexible FRP grids into 3DPC, we do not elaborate on the printability of the two concrete types, as they meet the 3DPC requirements.

For FC, the raw materials used in this study consisted of cementitious materials, fine aggregates, PE fibers, and chemical admixtures. The cementitious materials included P.O.42.5 R Portland cement (OPC), silica fume (SF), and fly ash (FA). Quartz sand (QS), with a maximum grain size of 0.18 mm, was used as the aggregate. HPMC served as a thixotropic agent to adjust the viscosity of FC, while PE fibers (1% volume fraction) were

incorporated for reinforcement, as detailed in Supplementary Table 2. To study the effect of HPMC addition on the hydration products of FC, additional groups were added for microscopic characterization analysis, as shown in Supplementary Table 3.

Fresh FC was prepared using a QJ-20 type mixer with a 20-liter capacity. The dry materials (OPC, FA, SF, and QS) were mixed at 140 rpm for 5 min, followed by the addition of water and HPMC, and mixing at 300 rpm for another 5 min. PE fibers were then added, and the mixture was agitated at 300 rpm for an additional 6 min. The FC pastes (without PE fibers and QS) were prepared with the same stirring process for the microscopic test.

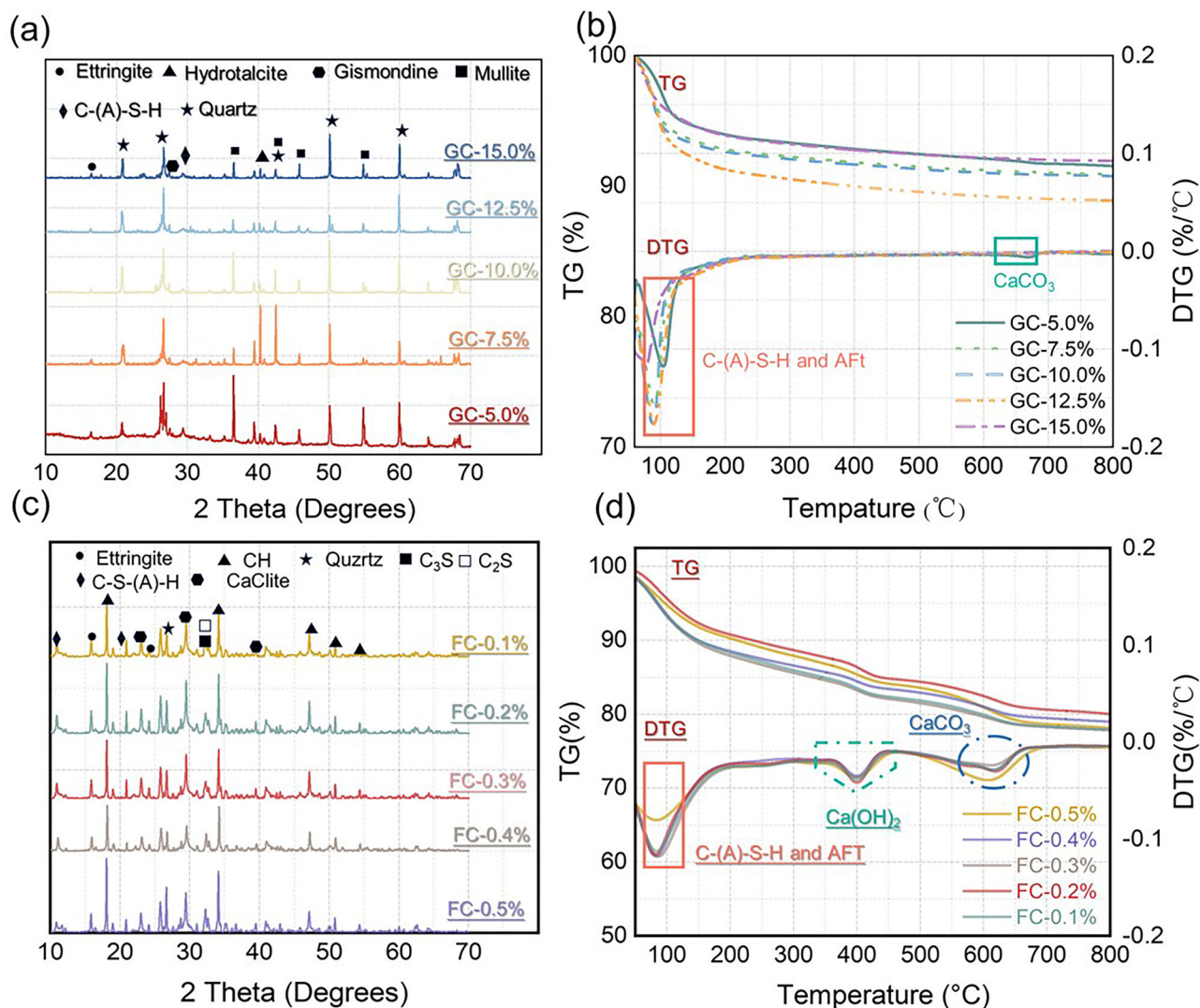


Fig. 4 | The influence of PSM and HPMC content on the hydration products of the printable GC and FC. a The XRD pattern of GC with different content of PSM at 28 d. **b** DTG-TG curves of GC at different PSM contents at 28 d. **c** The XRD pattern of FC with different content of HPMC at 28 d. **d** DTG-TG curves of FC at different

HPMC contents at 28 d. GC geopolymer concrete, FC fiber-reinforced concrete, HPMC hydroxypropyl methylcellulose, PSM sodium metasilicate, XRD X-ray diffraction, DTG-TG thermogravimetric analysis.

For GC, we used a one-part formulation (“just add water”), incorporating FA, SF, and ground granulated blast furnace slag (GGBFS) as cementitious materials. QS was used as the aggregate, and PSM acted as the alkali activator. We varied the PSM content for different groups, as shown in Supplementary Table 3. The preparation process involved mixing GGBFS, FA, SF, QS, and PSM before adding water. The mixture was stirred at 140 rpm for 2 min and 300 rpm for 5 min. The GC pastes (without QS) for microscopic tests followed the same stirring process.

In this research, we selected flexible FRP grids produced by Karbon Technology. As shown in Fig. 2a, the grid strips have a width of 4 mm, a thickness of 0.3 mm, and a center spacing of 20 mm between adjacent strips. According to tests conducted by the manufacturer, the elastic modulus, ultimate tensile strength, and ultimate tensile strain of the flexible FRP grids were 216.0 GPa, 2745.3 MPa, and 0.0126, respectively.

Mechanical tests

Three types of tests, including three-point bending tests, splitting tensile tests of the bond interface (the interface between two adjacent printing layers), and pull-out tests (based on the FRP-printed concrete bond joints) were conducted. A summary of these test setups is provided in Fig. 2.

Three-point bending tests

Seven 3DPC plates were tested under three-point bending. The plates measured 380 ± 2 mm in length, 120 ± 2 mm in width, and 52 ± 2 mm in height, with print filaments having a width of 20 mm and height of 13 mm. The plates tested under bending consisted of four layers of printed concrete, with the FC placed in the bottom two layers and the GC in the top two layers. This layering arrangement was designed intentionally to optimize structural efficiency, allowing each material to perform effectively according to its specific mechanical characteristics. Three types of flexible FRP grids were used, with widths of 20 mm (S1), 60 mm (S2), and 40 mm (S3), as shown in Fig. 2b. The grid configurations are illustrated in Fig. 2b, where blue rectangles represent the FRP grids. The plates were labeled according to the type and configuration of the flexible FRP grids; for example, S1-2R-6C refers to a plate with 2 rows and 6 columns of the S1-type FRP grids. All plates were cured at room temperature for 24 h, then placed in a standard curing room (humidity = 95%, temperature = 20 ± 2 °C) for 27 days. The displacement rate for the three-point bending tests was set at 0.2 mm/min. Mid-span displacement was measured using a linear variable differential transformer (LVDT), positioned as shown in Fig. 2b. Notably, S1-2R-6C

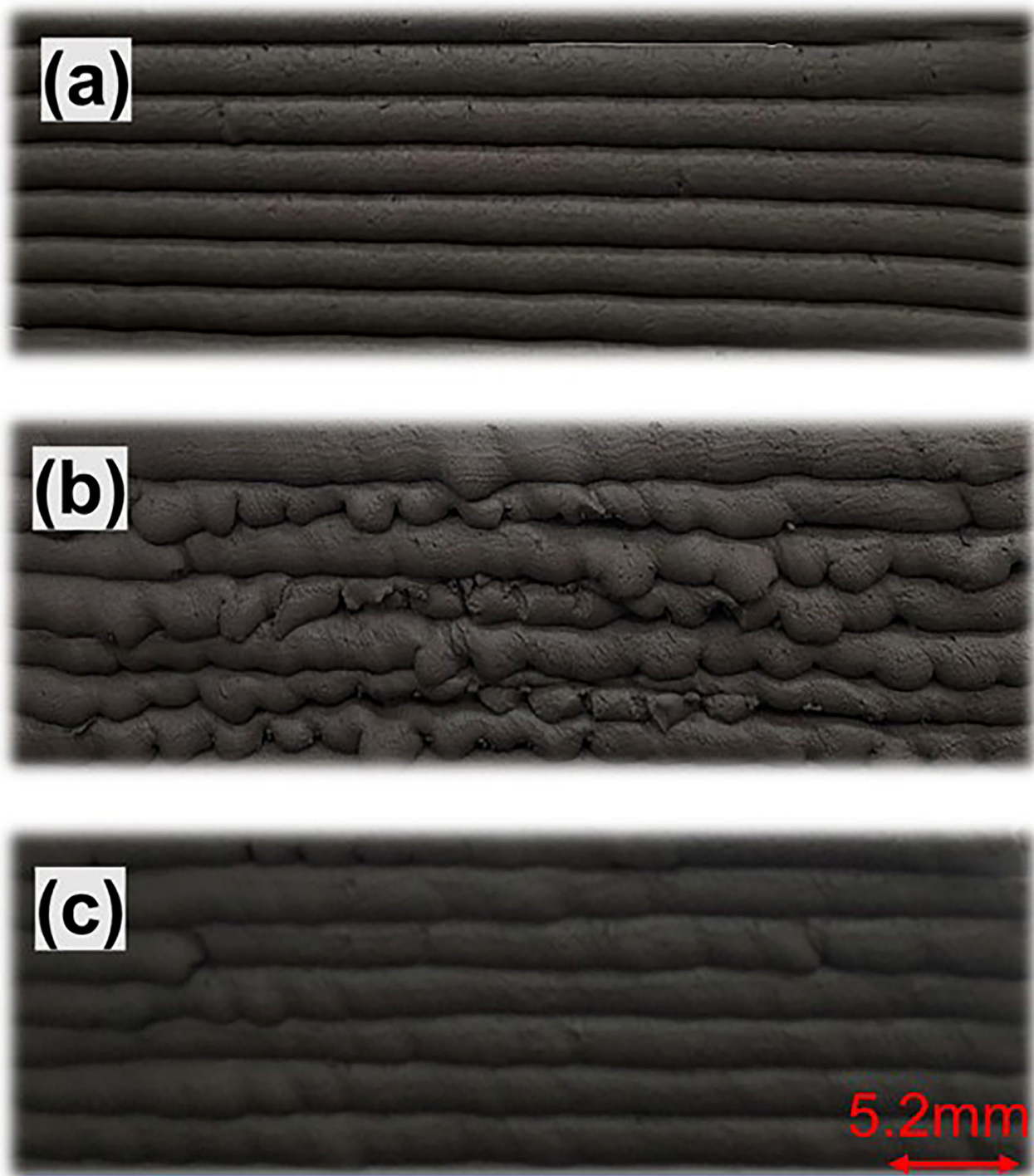


Fig. 5 | The surface morphology under different nozzle speed. **a** The surface morphology of concrete under 40 mm/s nozzle speed (without FRP grids). **b** The surface morphology of concrete under 40 mm/s nozzle speed (with FRP grids). **c** The surface morphology under 30 mm/s nozzle speed (with FRP grids).

had a cross-sectional reinforcement ratio of 1.14%, while the other plates had a ratio of 0.57%.

Splitting tensile tests of the interlayer interface

The splitting tensile tests of the bond interface were conducted using a test device designed by Liu, which included an upper and a lower fixture⁴¹. The samples for the splitting tensile tests had dimensions of 20 ± 2 mm in width, 100 ± 2 mm in length, and 26 ± 2 mm in height, as shown in Fig. 2d. The samples were cured at room temperature for 24 h and then transferred to a standard curing room for 27 days. The displacement rate for the tests was set

at 2 mm/min. To ensure test accuracy, each group of samples was tested in triplicate. The interlayer splitting tensile strength of the samples was measured using Eq. (1).

$$f_b = 2F/\pi A = 0.637F/A \quad (1)$$

where, f_b represents the interlayer interface splitting tensile strength (MPa); F represents the maximum load (N); A represents the area of failure plane (mm^2).

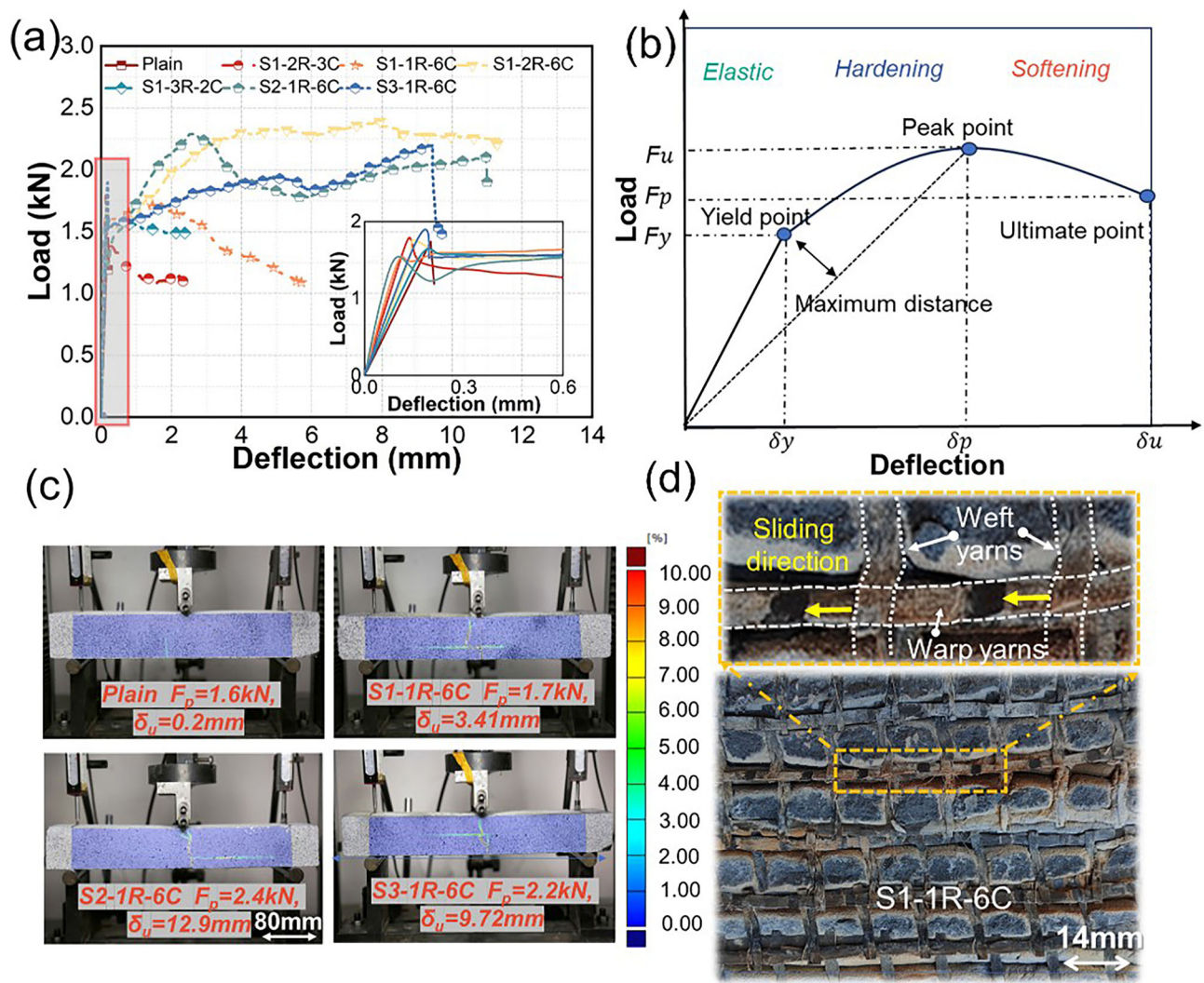


Fig. 6 | Results of three-point bending tests of 3DPC plates. a Load -deflection curves of test plates. **b** Loads correspond to yield and ultimate point. **c** Crack pattern of 3DPC plates at the ultimate point. **d** Morphology of the interlayer interface after three-point bending tests.

Pull-out tests

With reference to Akilesh Ramesh's previous study⁴² for pull-out tests, the device as shown in Fig. 2c⁴². The dimensions of samples were 30 ± 2 mm in length, 20 ± 2 mm in width, and 50 ± 2 mm in height. The bond joint samples had an FRP grid embedded length of 20 mm, a free length of 25 mm and a gripping length of 50 mm. The samples were cured at room temperature for 24 h and then transferred to a standard curing room for 27 days. The displacement rate for tests was 0.5 mm/min. The average bond strength between flexible FRP grids and 3DPC can be obtained based on Eq. (2).

$$\tau = \frac{P_{\max}}{n\pi dl} \quad (2)$$

where, $P_{\max}$ represents the peak load, n represents the amount of wrap yarns, d represents the equivalent bundle diameter and l represents the embedded length.

Microscopic characterization

Figure 3 presents the test s procedures for four microscopic characterization namely XRD, DTG-TG, MIP and μ CT. It should be particularly noted that XRD and DTG-TG were used to investigate the effect of PSM and HPMC addition on the hydration products of printable GC and FC, as shown in Supplementary Table 3. MIP and μ CT were used to analyze the changes in

the interlayer interface pore structure of 3DPC with the addition of FRP grids.

Hydration characteristics of GC and FC

The mineral phase evolution of GC and FC pastes under additives (PSM and HPMC) were analyzed using XRD ((Rigaku Dmax-2500PC, Japan). The instrument parameters were set from 10° to 70° , with a step size of 0.02° and a time of 0.01 s per step. Exploring the effect of PSM and PMC on the degree of hydration by DTG-TG. Simultaneous Thermal Analyzer (Mettler Toledo TGA/DSC 1/1600) was utilized for monitoring the weight loss of GC and FC while heating from 30°C to 1000°C ($10^\circ\text{C}/\text{min}$ of heating rate) and purging with N_2 gas ($100\text{ mL}/\text{min}$ of flow rate).

Microstructure analysis of the interlayer interface at FGC

MIP tests were performed using an automated mercury piezometer. Samples weighed 1–1.5 g and were removed from the interlayer of two layers by an electric drill. μ CT tests were conducted utilizing a high-resolution industrial CT scanner. The sample dimensions measured $26 \times 26 \times 26\text{ mm}^3$, with a scanning voltage of 190 kV, a current of 0.16 mA, and a resolution of $36.3\text{ }\mu\text{m}/\text{pixel}$. The resultant scanned images were subsequently analyzed using Avizo software to obtain specific pore parameters. Prior to testing, the samples of MIP and μ CT were immersed in anhydrous ethanol for 24 h and then heated in an oven at 45°C for at least 2 h until the weight of the samples no longer changed.

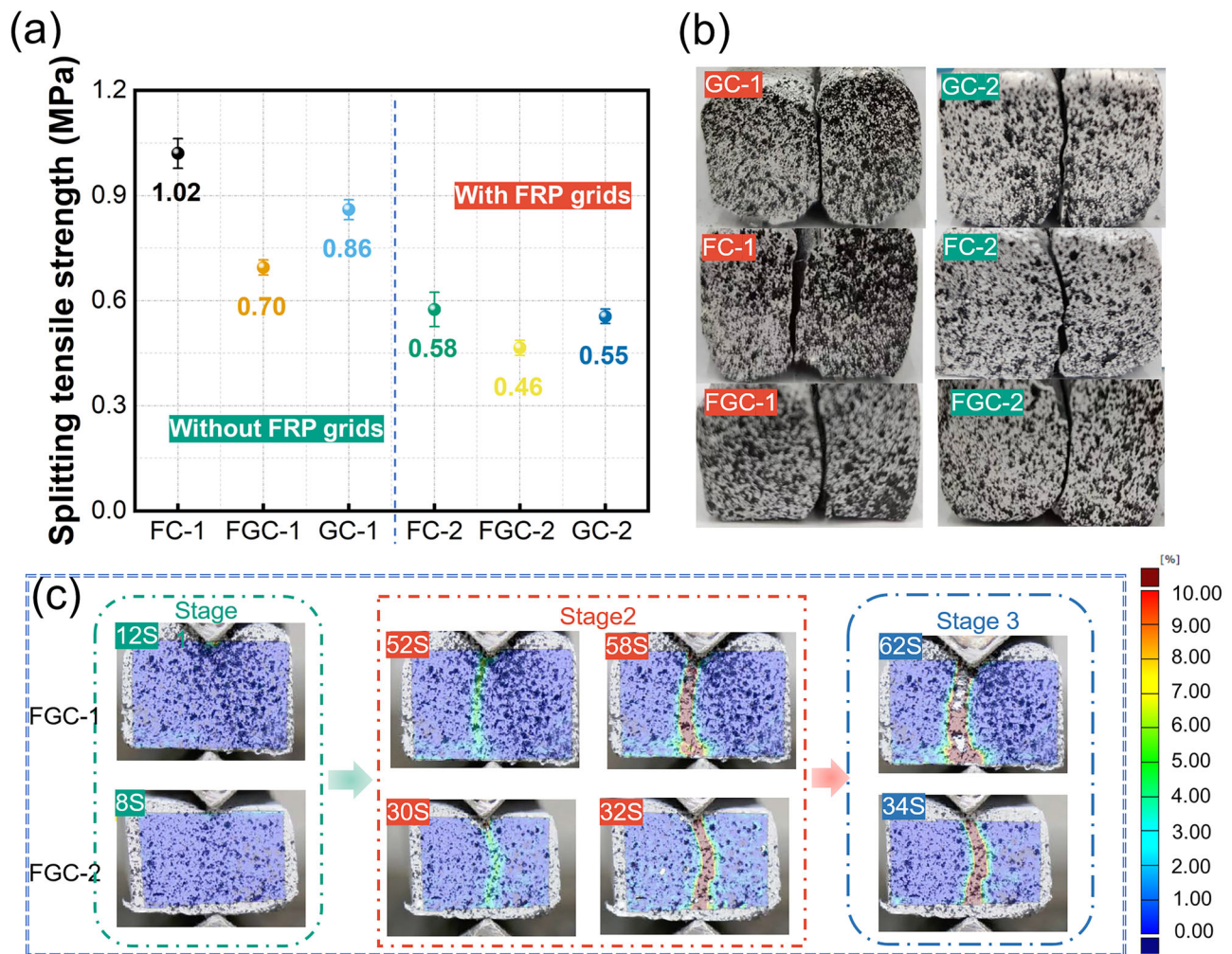


Fig. 7 | Results of Splitting tensile tests. **a** The splitting tensile strength of different interlayers (Data points represent the mean values from three independent experiments; error bars are defined as the mean $\pm$ standard deviation). **b** Interface cracking pattern after interlayer splitting tensile tests. **c** Interlayer interface strain of specimens (FC-1 represents fiber-reinforced concrete without FRP grid; GC-1

represents geopolymer concrete without FRP grid; FGC-1 represents functionally graded concrete without FRP grid; FC-2 represents fiber-reinforced concrete with FRP grid; GC-2 represents geopolymer concrete with FRP grid; FGC-2 represents functionally graded concrete with FRP grid).

Results and discussions

Hydration characteristics of printable GC and FC

Figure 4 shows the influence of PSM and HPMC content on the hydration products of the printable GC and FC. As shown in XRD pattern of GC paste (Fig. 4a), the hydration product composition remains largely unchanged with increasing PSM, primarily consisting of hydroxalite, calcium aluminate, carbonate minerals, and C-(A)-S-H, consistent with previous studies^{43,44}. A distinct quartz peak is observed, indicating unreacted FA residues. The quartz peak is most prominent in the GC-15.0% sample, suggesting over-alkalization at this PSM content, which encapsulated material particles with hydration products, limiting further hydration. This explains the weaker tobermorite-type C-(A)-S-H peak at 29° in the GC-15.0% sample. Mullite crystals are also present, but their intensity is reduced, reflecting the involvement of Al^{3+} in the reaction. A small peak near $2\theta = 10^\circ$ corresponds to ettringite (Aft), which strengthens the gel system. Hydroxalite, linked to slag content, is also identified⁴⁵. As revealed by XRD patterns in Fig. 4c, the main hydration products in FC paste consist of Aft, calcium hydroxide (CH), calcite (CaCO_3), and C-(A)-S-H gels. Notably, while the phase composition of hydration products remained largely unchanged with increasing HPMC content, the most pronounced CH diffraction peak observed in FC-0.5% suggests that HPMC at this dosage

may retard the hydration process, thereby reducing the consumption rate of CH.

Figure 4b, d presents the changes in the DTG-TG curves of GC and FC pastes with varying additive contents. As shown in the DTG-TG curves of Fig. 4b, a distinct mass loss peak observed at approximately 100°C corresponds to dehydration of both C-(A)-S-H and Aft phases⁴⁶. The characteristic decomposition peak of Aft overlaps with that of C-(A)-S-H, thus precluding separate quantification of these components. The intensity of this endothermic peak follows the sequence: GC-12.5% > GC-10.0% > GC-7.5% > GC-15.0% > GC-5.0%. The peak intensity initially increases and then decreases with increasing PSM content, with a noticeable reduction observed at a PSM content of 15%. This trend is consistent with the XRD results, which suggest that excessively high alkali concentrations impede the progress of the hydration reaction. As illustrated by the DTG-TG curves in Fig. 4d, the FC-0.5% paste exhibited a noticeably lower endothermic peak intensity for C-(A)-S-H gels and Aft compared to other pastes, indicating that the HPMC dosage of 0.5% notably inhibited the hydration process. This observation aligns with the XRD results, confirming the retarding effect of HPMC. Based on the combined analysis of XRD and DTG-TG data, FC and GC composites with 0.1% HPMC and 12.5% PSM were selected for subsequent 3D printing performance evaluation.

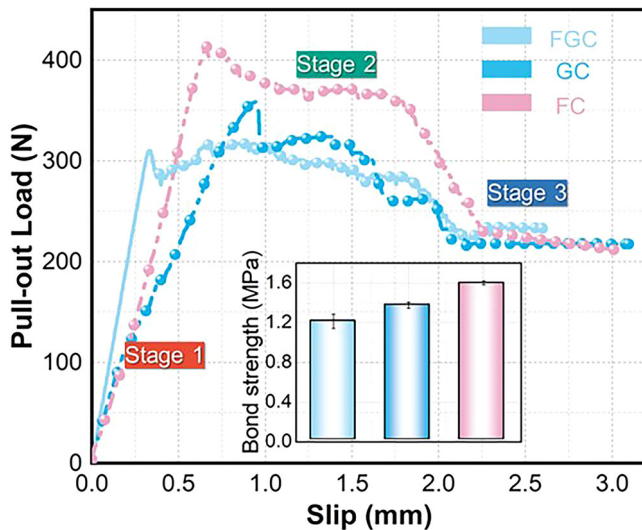


Fig. 8 | Pull-out load vs slip curves: Stage 1 represents sharp linear stage; Stage 2 represents descending stage; Stage 3 represents steady flat stage. Data points represent the mean values from three independent experiments; error bars are defined as the mean $\pm$ standard deviation; GC geopolymer concrete, FC fiber-reinforced concrete, FGC functionally graded concrete.

The nozzle movement speed

Figure 5 characterizes the surface morphology of printed filament under varying nozzle speeds. Comparative analysis reveals that at identical printing velocities, the incorporation of FRP grids induces marked surface deformation in 3DPC (Fig. 5a, b), attributable to localized stress concentration. Reducing the nozzle speed from 40 to 30 mm/s, effectively enhancing coordinated deformation capacity. The 30 mm/s parameter was therefore selected for subsequent printing based on optimization of surface integrity.

Mechanical test results

Three-point bending test results. Results of three-point bending tests of 3DPC plates are presented in Fig. 6. As shown in load-deflection curves of 3DPC plates (see Fig. 6a), the curves for flexible FRP grid-reinforced 3DPC plates can be divided into either two-portion (an elastic portion and a softening portion) curves or three-portion (an elastic portion, a hardening portion, and a softening portion) curves. The effectively reinforced plates exhibit a three-portion behavior, as shown in Fig. 6b. In contrast, the plain plate exhibits only an elastic stage and experienced damage as soon as cracks appeared, which is attributed to the inherent brittleness of concrete^{13,39,47,48}.

Flexible FRP grid-reinforced 3DPC plates show higher load-bearing capacity and deflection than plain plates. Figure 6a shows the S1-2R-6C configuration with the highest capacity and deflection, due to its higher reinforcement ratio of 1.14%. Plates with a 0.57% reinforcement ratio, like S2-1R-6C, achieve the highest peak load. S1-2R-3C has 75% of the load and 21.7% of the deflection of S1-2R-6C, but its peak load (1.8 kN) and deflection (2.5 mm) represent increases of 0.13 and 11.5 times over the plain plate. Overall, load-bearing capacity and deflection increase with more flexible FRP grids, especially deflection. However, increasing grid count has little effect on cracking load, which ranges from 1.6 to 1.8 kN. Variations in performance for similar configurations are due to changes in effective depth and grid placement^{18,49}.

At the ultimate point (load reduced to 80% of peak, see Fig. 6a), S2-1R-6C and S1-2R-6C have peak loads and deflections of 2.3 kN and 12.5 mm and 2.4 kN and 12.9 mm, respectively. Halving the flexible FRP grid width does not noticeably change these values, even though S1-2R-6C has twice the reinforcement ratio of S2-1R-6C. With reasonable reinforcement ratio, a larger grid width results in more significant reinforcement effectiveness.

Therefore, carefully adjusting the width and configuration ratio of FRP grids is essential to optimize the load-bearing capacity and ductility of 3DPC plates.

Figure 6c shows the crack patterns of 3DPC plates at the ultimate point. Plain 3DPC plates develop rapid cracks at the bottom, leading to typical brittle failure^{13,39,47,48}. In contrast, FRP grid-reinforced 3DPC plates exhibit a single crack similar to those in 3DPC reinforced with steel cables²⁹. Some interlayer cracks along the FRP grids are seen in the S1-1R-6C, S2-1R-6C, and S3-1R-6C specimens, but these do not noticeably affect the plates' load-bearing capacity¹⁶.

Figure 6d shows the interlayer interface morphology, revealing noticeable slippage between the flexible FRP grids and the concrete matrix. This slippage caused the single-crack failure in the FRP grid-reinforced 3DPC plates. Notably, only the warp yarns slipped while the weft yarns remained fixed. This differential behavior stems from weak interfacial bonding at warp yarn/weft yarn junctions, combined with intrinsic structural effects: Resin-rich zones at fiber crossover points exhibit localized stress concentrations, triggering interfacial debonding under tension. Consequently, stress transfer occurs primarily through weft yarns anchored in concrete, forcing warp yarns to slide along their longitudinal axis.

Splitting tensile tests of the interlayer interface test results. Results of splitting tensile tests of interlayer in 3DPC plates are presented in Fig. 7. Figure 7a shows the splitting tensile strength of six interlayer types in 3DPC plates: FC-1, FGC-1, GC-1 (without flexible FRP grids), and FC-2, FGC-2, GC-2 (with flexible FRP grids). FC-1 exhibited the highest strength, while FGC-1 had the lowest. Fiber addition improved tensile strength by bridging interfaces and aligning with the printing direction⁵⁰. However, flexible FRP grids reduced strength by 43.6%, 33.1%, and 35.5% for FC-2, FGC-2, and GC-2, respectively, likely due to decreased contact area between layers. Further microstructural analysis is needed to explain this reduction.

When printing the same type of concrete, an noticeable amount of free water is present on the surface of the filaments, promoting a strong bond between the concrete layers under gravity. However, in the FGC interlayer, the need to switch between different concretes results in a longer interval time between layers, leading to the evaporation of a substantial amount of free water^{41,51,52}. This increases the porosity and reduces the interlayer splitting tensile strength. Additionally, the initial setting time of both concretes in this study is within 25 min. If the interval time exceeds this setting time, the underlying concrete begins to hydrate, forming a cold joint between the two concretes, which further weakens the interlayer splitting tensile strength^{51,52}.

Figure 7b shows the interface crack patterns after interlayer splitting tensile tests. Groups without flexible FRP grids had straight, minimally fluctuating cracks, while those with FRP grids exhibited wider fluctuations and irregular paths, bulging toward the upper print filaments. This indicates that flexible FRP grids alter the force transfer path at the interlayer interface. This indicates that flexible FRP grids alter the force transfer path at the interlayer interface. The underlying cause is attributed to the grid's introduction, which noticeably elevates interlayer porosity and intensifies pore distribution heterogeneity within the region (For detailed explanation, please refer to the microstructure characterization). This highly heterogeneous pore structure governs the crack propagation path during splitting failure: fractures are forced to zigzag through spatially localized, high-porosity weak zones with low strength. To connect these areas or seek minimal stress-transfer pathways between them, cracks develop highly fluctuating and upward-tilted failure trajectories.

Digital image correlation (DIC) analysis of typical FGC-1 and FGC-2 specimens (see Fig. 7c) revealed three distinct stages of interlayer crack-ing: Stage 1: Stress concentrates at the loading indenter tip, with a gradual increase in strain. Stage 2: A strain-concentrating band forms at the interlayer interface without immediate failure, maintaining the stress state briefly. Stage 3: Crack width widens, leading to sample rupture. The fracture surface lifted from the bottom filaments, and the crack

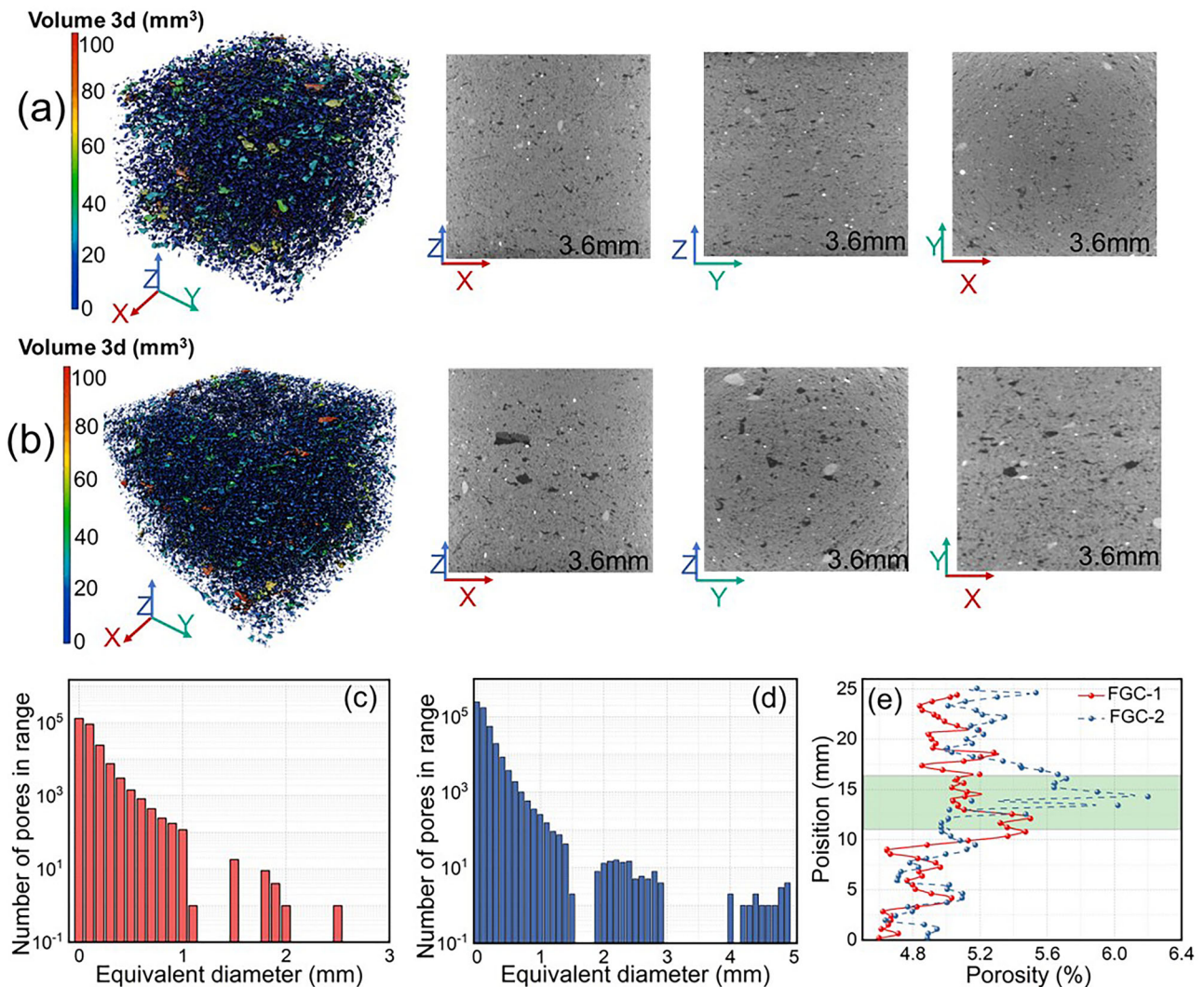


Fig. 9 | Pore structure of interlayer interface at different concrete. Rendered 3D model and μ CT scanned image of **a** FGC-1 and **b** FGC-2; Number of pores in **c** FGC-1 and **d** FGC-2; **e** The height-wise distribution of pores in the FGC-1 and FGC-2

(FGC-1 represents functionally graded concrete without FRP grid; FGC-2 represents functionally graded concrete with FRP grid).

path curved, indicating incomplete interlayer bonding and internal interface defects.

Pull-out test results. Figure 8 shows the load-slip curves for different concrete sample under pull-out load, all exhibiting three stages: (1) Sharp linear stage: Indicates elongation of the flexible FRP grids. (2) Descending stage: Reflects debonding between the FRP grids and the concrete matrix. (3) Steady flat stage: Represents frictional slipping. Only the warp yarns were pulled out during the tests, while the weft yarns remained embedded in the concrete, consistent with previous studies^{42,53}. No rupture of the FRP grids or visible concrete damage was observed.

Figure 8 shows that FC has the highest average bond strength at 1.6 MPa, which is 16.7% and 32.2% higher than GC and FGC, respectively. There is currently no standardized method for evaluating bond strength between flexible FRP grids and 3DPC. A previous study⁴² reported bond strengths ranging from 1.14 to 2.76 MPa. The lower bond strength in this study may be due to placing the FRP grid printer head below the concrete printer head, creating gaps between the grids and concrete. Further microstructural analysis is conducted to investigate the decrease in bond strength between layers. It must be emphasized that the elevated interlayer porosity not only diminished the bond strength but also compromised the stress transfer efficiency between the FRP grid and the concrete matrix. This

inefficiency in force redistribution resulted in a single-crack failure mode in the FGC plates during flexural testing.

Microstructure characterization

Internal voids noticeably impact the mechanical properties of 3DPC. Air trapped between printed layers forms cavities, altering interlayer interface bonding⁵⁴ and causing anisotropy and reduced strength compared to mould-cast concrete^{54,55}. 3DPC pores are predominantly elongated, especially the interlayer pores, due to printhead movement during printing. This elongated morphology is further highlighted in the rendered 3D model.

Figure 9c, d quantify pores in FGC-1 and FGC-2 using equivalent diameter to represent spherical particles with similar properties^{7,54}. Both specimens show a normal distribution of pore sizes, with similar numbers of pores <1 mm. However, FGC-2 has noticeably more pores >1 mm, which greatly impact bonding performance⁵⁴.

Figure 9e shows the vertical pore distribution in FGC-1 and FGC-2. The addition of flexible FRP grids noticeably increases interlayer porosity,

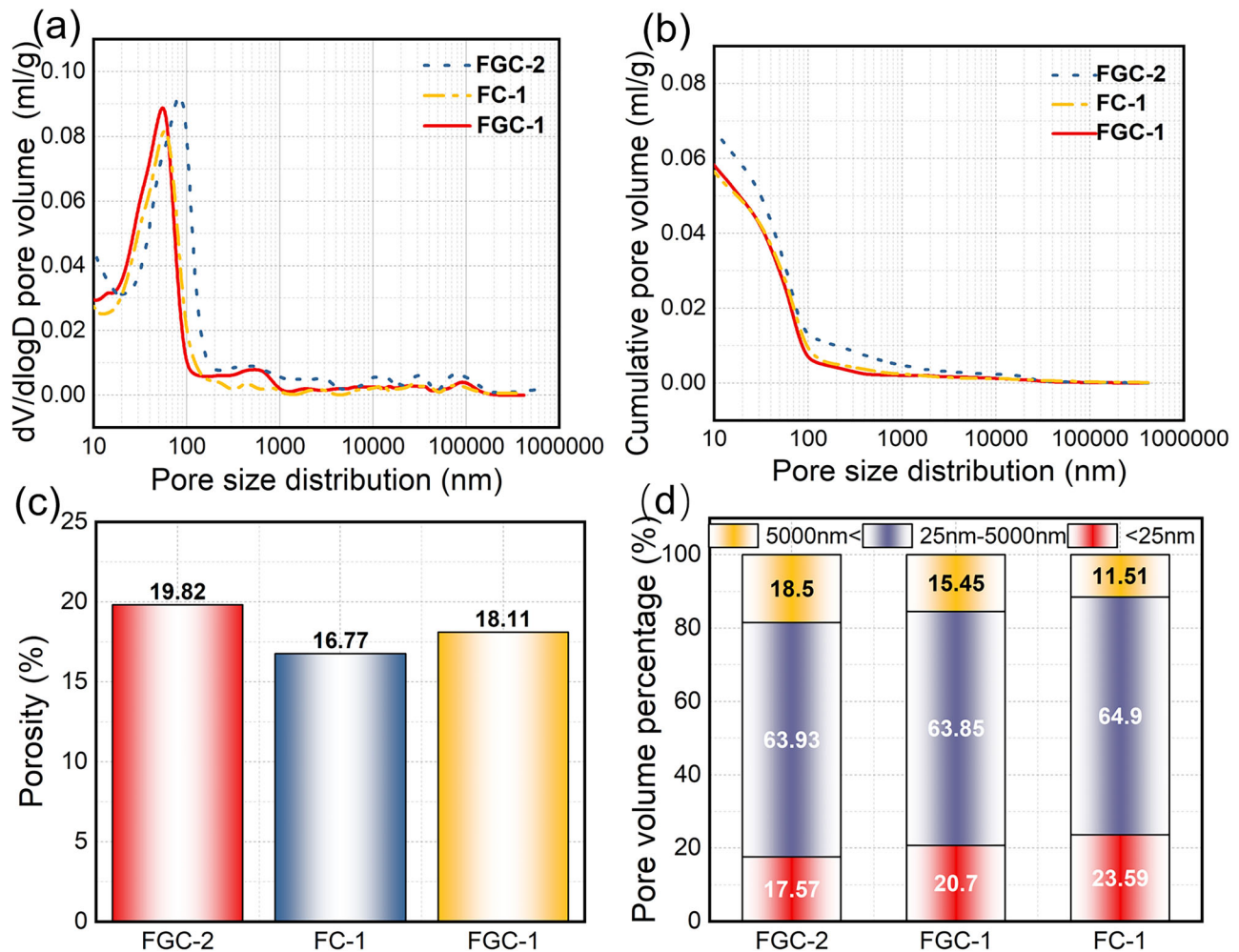


Fig. 10 | Pore structure of interlayer interface at different concrete. **a** Differential curve of pore size distribution; **b** Cumulative pore volume distribution; **c** Porosity; **d** Pore volume ratio (FC-1 represents functionally graded concrete without FRP

grid; FGC-2 represents functionally graded concrete with FRP grid; FC-1 represents fiber-reinforced concrete without FRP grid).

especially for pores >1 mm. Additionally, the bottom concrete layer has lower porosity than the upper layer, due to gravitational compression from the top layer.

μ CT is utilized to measure pore sizes greater than three times the voxel size used, while smaller pores, down to the nanoscale, can be measured using MIP tests. Figure 10 shows the pore structure from MIP testing, where the FC-1 specimen had the lowest interlayer and cumulative porosity. In contrast, FGC-1 and FGC-2 exhibited much higher porosity and larger pore sizes, primarily due to reduced interlayer water content from longer layer intervals^{41,42,51}. Concrete porosity positively correlated with strength, consistent with the interlayer splitting tensile strength results⁵⁵. According to the previous study⁵⁶, interlayer porosity is classified as <25 nm, 25–5000 nm, and >5000 nm. FGC-1 and FGC-2 showed increased porosity, especially pores >5000 nm, after adding flexible FRP grids. This increase in larger pores aligns with μ CT test results and partially explains the reduced interlayer bond strength after adding flexible FRP grids. The high yield strength requirement of 3DPC and the use of dual nozzles for simultaneous printing likely caused voids beneath the FRP grids, increasing porosity. This is similar to the previous study²⁸, which used steel cables for simultaneous printing.

Test results from MIP and CT reveal that embedding FRP grids noticeably increases the interlayer porosity of 3DPC, leading to a substantial degradation in the FRP-concrete interfacial bond strength. This weakened bond induces interfacial slip of the FRP grid at crack locations under flexural loading, which hinders stress transfer to adjacent concrete regions and suppresses the formation of new cracks. Consequently, all deformations are

concentrated within a single dominant crack, triggering a brittle single-crack failure pattern as opposed to a ductile multi-crack development.

Conclusions

In this study, we investigated the feasibility of a novel dual-nozzle in-process embedded technology for simultaneously printing flexible FRP grids into 3DPC elements. We conducted three mechanical tests: three-point bending tests, interlayer splitting tensile tests, and pull-out tests. From these tests, we drew the following conclusions:

Enhanced load-bearing capacity and deflection: Incorporating FRP grids noticeably increased the load-bearing capacity and deflection of 3D-printed FGC plates generally increases with the reinforcement ratio. The S1-2R-6C plate exhibited a load-bearing capacity and deflection 1.41 and 55.2 times greater than the plain plate, respectively. Additionally, S2-1R-6C and S1-2R-6C showed similar deflection and load-bearing capacity, despite S1-2R-6C having double the reinforcement ratio of S2-1R-6C. This indicates that the width of a single flexible FRP grid noticeably impacts both load-bearing capacity and deflection.

Several challenges were identified: The FRP grids altered the force distribution at the layer interfaces and reduced the contact area between adjacent concrete layers, resulting in decreased interlayer splitting strength. Compared to FC-1, the interlayer splitting strength of FGC-1, GC-1, FC-2, FGC-2, and GC-2 decreased by 43.6%, 33.1%, and 35.5%, respectively. Concurrently, the addition of FRP grids elevated interlayer porosity levels and noticeably increased the population of elongated pores exceeding 1 mm

in size—a phenomenon directly correlated with the observed decline in interfacial bond performance. This weakened bonding induces interfacial slip of the FRP grid at crack locations under flexural loading, which hinders stress transfer to adjacent concrete regions and suppresses the formation of new cracks, resulting in only a single crack upon failure of the 3DPC plate. In addition, the increased porosity alters interlayer force transmission pathways, leading to noticeably different crack propagation patterns during splitting failure. Load-slip analysis further revealed consistent behavioral patterns across all samples, characterized by three distinct phases: a sharp linear stage, followed by a descending stage, and ultimately a steady flat stage. Within this framework, FC samples demonstrated highest bond strength, reaching 1.6 MPa and exceeding GC and FGC specimens by 16.7% and 32.2% respectively, though these values remain modest against the previous study. Most critically, the insufficient bonding integrity between warp and weft yarns within the FRP grids precipitated slippage between warp yarns and the concrete matrix during loading, thereby constraining the full reinforcement potential of the composite system.

Future research: To enhance reinforcement, we need to strengthen the bond between warp and weft yarns. Because inadequate inter-yarn bonding between warp and weft filaments in the FRP grid was identified as a structural deficiency. This combination triggers interfacial slip between warp yarns and concrete under loading, resulting in reinforcement effectiveness that underperforms manually placed steel rebars and FRP systems. Additionally, reducing voids formed during the printing process is crucial, as they negatively affect the durability of 3DPC components. This could be achieved by applying expanding agents to the FRP grids to minimize the impact of voids on the service life of the structures while ensuring robust warp and weft yarns bonding to preclude interfacial slippage. Furthermore, although this device facilitates in-plane reinforcement placement, orthogonal reinforcement across layer interfaces remains unresolved. Next-generation systems should integrate: (a) Z-direction fiber insertion nozzles, and (b) printable anisotropic reinforcements (e.g., directional steel fibers).

Data availability

Data will be made available on request.

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Competing interests

The authors declare no competing interests.

Additional information

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